



# **Chapter 4 : Intermediate SQL**

**Database Principle and Design**



# Outline

- Join Expressions
- Views
- Transactions
- Integrity Constraints
- SQL Data Types and Schemas
- Index Definition in SQL
- Authorization



## 4.1 Join Operations

In SQL, there are two ways to combine information from multiple relations:

- Use Cartesian product operation (list tables in **from** clause)
  - Example: List the names of instructors along with the course ID of the courses that they teach  
**select** *name, course\_id* **from** *instructor, teaches*  
**where** *instructor.ID = teaches.ID*;
- Write **join** operations in **from** clause
  - Example:  
**select** *name, course\_id* **from** *instructor join teaches*  
**on** *instructor.ID = teaches.ID*
- The **on** condition above specifies that a tuple from *instructor* matches a tuple from *teaches* if their *ID* values are equal.
- The **on** condition is written like a **where** clause predicate except for the use of the keyword **on**



## 4.1 Join Operations

In SQL, there are two ways to combine information from multiple relations:

- Write **join** operations in **from** clause

**select** *name, course\_id* **from** *instructor* **join** *teaches*  
**on** *instructor.ID = teaches.ID*

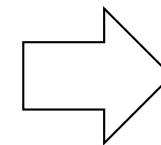
- The **on** condition above specifies that a tuple from *instructor* matches a tuple from *teaches* if their *ID* values are equal.
- The **on** condition is written like a **where** clause predicate except for the use of the keyword **on**

*instructor* relation

| <i>ID</i> | <i>name</i> | <i>dept_name</i> | <i>salary</i> |
|-----------|-------------|------------------|---------------|
| 10101     | Srinivasan  | Comp. Sci.       | 65000         |
| 12121     | Wu          | Finance          |               |
| 15151     | Mozart      | Music            |               |
| 22222     | Einstein    | Physics          |               |
| 32343     | El Said     | History          |               |
| 33456     | Gold        | Physics          |               |
| 45565     | Katz        | Comp. Sci.       |               |
| 58583     | Califieri   | History          |               |
| 76543     | Singh       | Finance          |               |
| 76766     | Crick       | Biology          |               |
| 83821     | Brandt      | Comp. Sci.       |               |
| 98345     | Kim         | Elec. Eng.       |               |

*teaches* relation

| <i>ID</i> | <i>course_id</i> | <i>sec_id</i> | <i>semester</i> | <i>year</i> |
|-----------|------------------|---------------|-----------------|-------------|
| 10101     | CS-101           | 1             | Fall            | 2017        |
| 10101     | CS-315           | 1             | Spring          | 2018        |
| 10101     | CS-347           | 1             | Fall            | 2017        |
| 12121     | FIN-201          | 1             | Spring          | 2018        |
| 15151     | MU-199           | 1             | Spring          | 2018        |
| 22222     | PHY-101          | 1             | Fall            | 2017        |
| 32343     | HIS-351          | 1             | Spring          | 2018        |
| 45565     | CS-101           | 1             | Spring          | 2018        |
| 45565     | CS-319           | 1             | Spring          | 2018        |
| 76766     | BIO-101          | 1             | Summer          | 2017        |
| 76766     | BIO-301          | 1             | Summer          | 2018        |
| 83821     | CS-190           | 1             | Spring          | 2017        |
| 83821     | CS-190           | 2             | Spring          | 2017        |
| 83821     | CS-319           | 2             | Spring          | 2018        |
| 98345     | EE-181           | 1             | Spring          | 2017        |



| <i>name</i> | <i>course_id</i> |
|-------------|------------------|
| Srinivasan  | CS-101           |
| Srinivasan  | CS-315           |
| Srinivasan  | CS-347           |
| Wu          | FIN-201          |
| Mozart      | MU-199           |
| Einstein    | PHY-101          |
| El Said     | HIS-351          |
| Katz        | CS-101           |



## 4.1 Join Operations (Cont.)

- A **from** clause can have multiple relations combined using **join** operations, for example

**select** *name, title* **from** *instructor* **join** *teaches* **on** *instructor.ID = teaches.ID*  
**join** *course* **on** *teaches.course\_id = course.course\_id*

- Two types of **join** operations:
  - Inner join
  - Outer join

| <i>ID</i> | <i>name</i> | <i>dept_name</i> | <i>salary</i> |
|-----------|-------------|------------------|---------------|
| 10101     | Srinivasan  | Comp. Sci.       | 65000         |
| 12121     | Wu          | Finance          |               |
| 15151     | Mozart      | Music            |               |
| 22222     | Einstein    | Physics          |               |
| 32343     | El Said     | History          |               |
| 33456     | Gold        | Physics          |               |
| 45565     | Katz        | Comp. Sci.       |               |
| 58583     | Califieri   | History          |               |
| 76543     | Singh       | Finance          |               |
| 76766     | Crick       | Biology          |               |
| 83821     | Brandt      | Comp. Sci.       |               |
| 98345     | Kim         | Electrical       |               |

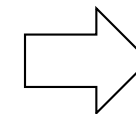
*instructor* relation

*teaches* relation

| <i>ID</i> | <i>course_id</i> | <i>sec_id</i> | <i>semester</i> | <i>year</i> |
|-----------|------------------|---------------|-----------------|-------------|
| 10101     | CS-101           | 1             | Fall            | 2017        |
| 10101     | CS-315           |               |                 |             |
| 10101     | CS-347           |               |                 |             |
| 12121     | FIN-201          |               |                 |             |
| 15151     | MU-199           |               |                 |             |
| 22222     | PHY-101          |               |                 |             |
| 32343     | HIS-351          |               |                 |             |
| 45565     | CS-101           |               |                 |             |
| 45565     | CS-319           |               |                 |             |
| 76766     | BIO-101          |               |                 |             |
| 76766     | BIO-301          |               |                 |             |
| 83821     | CS-190           |               |                 |             |
| 83821     | CS-190           |               |                 |             |
| 83821     | CS-319           |               |                 |             |
| 98345     | EE-181           |               |                 |             |

*course* relation

| <i>course_id</i> | <i>title</i>               | <i>dept_name</i> | <i>credits</i> |
|------------------|----------------------------|------------------|----------------|
| BIO-101          | Intro. to Biology          | Biology          | 4              |
| BIO-301          | Genetics                   | Biology          | 4              |
| BIO-399          | Computational Biology      | Biology          | 3              |
| CS-101           | Intro. to Computer Science | Comp. Sci.       | 4              |
| CS-190           | Game Design                | Comp. Sci.       | 4              |
| CS-315           | Robotics                   | Comp. Sci.       | 3              |
| CS-319           | Image Processing           | Comp. Sci.       | 3              |
| CS-347           | Database System Concepts   | Comp. Sci.       | 3              |
| EE-181           | Intro. to Digital Systems  | Elec. Eng.       | 3              |
| FIN-201          | Investment Banking         | Finance          | 3              |
| HIS-351          | World History              | History          | 3              |
| MU-199           | Music Video Production     | Music            | 3              |
| PHY-101          | Physical Principles        | Physics          | 4              |



| <i>name</i> | <i>title</i>               |
|-------------|----------------------------|
| Srinivasan  | Intro. to Computer Science |
| Srinivasan  | Robotics                   |
| Srinivasan  | Database System Concepts   |
| Wu          | Investment Banking         |
| Mozart      | Music Video Production     |
| Einstein    | Physical Principles        |
| El Said     | World History              |
| Katz        | Intro. to Computer Science |
| Katz        | Image Processing           |



## 4.1.1 Inner Join

- The result of **inner join** contains only **matching tuples**.
- Query example: List students along with the sections that they take  
**select \* from student inner join takes on student.ID = takes.ID**
- **inner** is optional
- Equivalent to:  
**select \* from student , takes where student.ID = takes.ID**
- A join operation **must** have a join condition. So the above join query can not be written as following

**select \* from student inner join takes where student.ID = takes.ID**

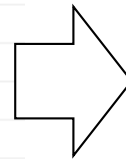


| ID    | name     | dept_name  | tot_cred |
|-------|----------|------------|----------|
| 00128 | Zhang    | Comp. Sci. | 102      |
| 12345 | Shankar  | Comp. Sci. | 32       |
| 19991 | Brandt   | History    | 80       |
| 23121 | Chavez   | Finance    | 110      |
| 44553 | Peltier  | Physics    | 56       |
| 45678 | Levy     | Physics    | 46       |
| 54321 | Williams | Comp. Sci. | 54       |
| 55739 | Sanchez  | Comp. Sci. | 54       |
| 70557 | Snow     | Comp. Sci. | 54       |
| 76543 | Brown    | Comp. Sci. | 54       |
| 76653 | Aoi      | Comp. Sci. | 54       |
| 98765 | Bourikas | Comp. Sci. | 54       |
| 98988 | Tanaka   | Comp. Sci. | 54       |

*student*

*takes*

| ID    | course... | sec_id | semest... | year | grade |
|-------|-----------|--------|-----------|------|-------|
| 00128 | CS-101    | 1      | Fall      | 2017 | A     |
| 00128 | CS-347    | 1      | Fall      | 2017 | A-    |
| 12345 | CS-101    | 1      | Fall      | 2017 | C     |
| 12345 | CS-190    | 2      | Spring    | 2017 | A     |
| 12345 | CS-315    | 1      | Spring    | 2018 | A     |
| 12345 | CS-347    | 1      | Fall      | 2017 | A     |
| 19991 | HIS-351   | 1      | Spring    | 2018 | B     |
| 23121 | FIN-201   | 1      | Spring    | 2018 | C+    |
| 44553 | PHY-101   | 1      | Fall      | 2017 | B-    |
| 45678 | CS-101    | 1      | Fall      | 2017 | F     |
| 45678 | CS-101    | 1      | Spring    | 2018 | B+    |
| 45678 | CS-319    | 1      | Spring    | 2018 | B     |
| 54321 | CS-101    | 1      | Fall      | 2017 | A-    |
| 54321 | CS-190    | 2      | Spring    | 2017 | B+    |



| ID    | name     | dept_name  | tot_cr... | ID    | course... | sec_id | semest... | year | grade |
|-------|----------|------------|-----------|-------|-----------|--------|-----------|------|-------|
| 00128 | Zhang    | Comp. Sci. | 102       | 00128 | CS-101    | 1      | Fall      | 2017 | A     |
| 00128 | Zhang    | Comp. Sci. | 102       | 00128 | CS-347    | 1      | Fall      | 2017 | A-    |
| 12345 | Shankar  | Comp. Sci. | 32        | 12345 | CS-101    | 1      | Fall      | 2017 | C     |
| 12345 | Shankar  | Comp. Sci. | 32        | 12345 | CS-190    | 2      | Spring    | 2017 | A     |
| 12345 | Shankar  | Comp. Sci. | 32        | 12345 | CS-315    | 1      | Spring    | 2018 | A     |
| 12345 | Shankar  | Comp. Sci. | 32        | 12345 | CS-347    | 1      | Fall      | 2017 | A     |
| 19991 | Brandt   | History    | 80        | 19991 | HIS-351   | 1      | Spring    | 2018 | B     |
| 23121 | Chavez   | Finance    | 110       | 23121 | FIN-201   | 1      | Spring    | 2018 | C+    |
| 44553 | Peltier  | Physics    | 56        | 44553 | PHY-101   | 1      | Fall      | 2017 | B-    |
| 45678 | Levy     | Physics    | 46        | 45678 | CS-101    | 1      | Fall      | 2017 | F     |
| 45678 | Levy     | Physics    | 46        | 45678 | CS-101    | 1      | Spring    | 2018 | B+    |
| 45678 | Levy     | Physics    | 46        | 45678 | CS-319    | 1      | Spring    | 2018 | B     |
| 54321 | Williams | Comp. Sci. | 54        | 54321 | CS-101    | 1      | Fall      | 2017 | A-    |
| 54321 | Williams | Comp. Sci. | 54        | 54321 | CS-190    | 2      | Spring    | 2017 | B+    |

*result*



## 4.1.2 Outer Join

- An extension of the inner join operation that avoids loss of information.
- Computes the inner join and then adds tuples from one relation that does not match tuples in the other relation to the result of the **join**.
- Three forms of outer join:
  - **left outer join**
  - **right outer join**
  - **full outer join**
- **outer** is optional





# Left Outer Join

- Computes the inner join and then adds tuples from the left relation that does not match tuples in the right relation to the result of the join.
- Example: List **all** students along with the sections that they take  
**select \* from student left join takes on student.ID = takes.ID**
- Example: List **all** students in the Physics department along with the sections that they take

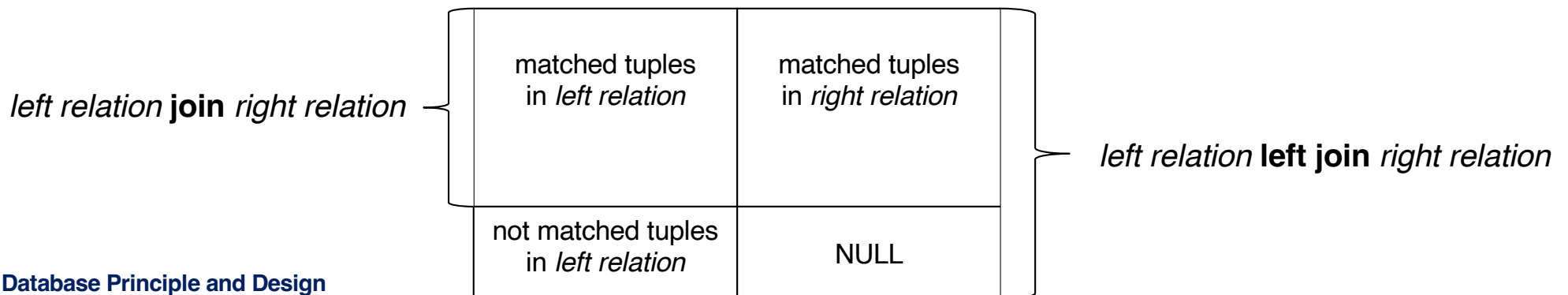
**select \* from student left join takes on student.ID = takes.ID  
where dept\_name = 'Physics'**

The above query can not be written as following

**select \* from student left join takes on student.ID = takes.ID  
and dept\_name = 'Physics'**



- Example: Find students who did not take any sections.  
**select student.\* from student left join takes on student.ID = takes.ID  
where takes.ID is null**







# Left Outer Join (Cont.)

- List **all** course titles and their prerequisite course titles (if any)

```
select c1.title as course, c2.title as prerequisite
from course as c1 left join prereq as p on c1.course_id=p.course_id
left join course as c2 on c2.course_id=p.prereq_id
```

| course_id | title                      | dept_name  | credits |
|-----------|----------------------------|------------|---------|
| BIO-101   | Intro. to Biology          | Biology    | 4       |
| BIO-301   | Genetics                   | Biology    | 4       |
| BIO-399   | Computational Biology      | Biology    | 3       |
| CS-101    | Intro. to Computer Science | Comp. Sci. | 4       |
| CS-190    | Game Design                | Comp. Sci. | 4       |
| CS-315    | Robotics                   | Comp. Sci. | 3       |
| CS-319    | Image Processing           |            |         |
| CS-347    | Database System Concepts   |            |         |
| EE-181    | Intro. to Digital Systems  |            |         |
| FIN-201   | Investment Banking         |            |         |
| HIS-351   | World History              |            |         |
| MU-199    | Music Video Production     |            |         |
| PHY-101   | Physical Principles        |            |         |

*course*

| course_id | prereq_id |
|-----------|-----------|
| BIO-301   | BIO-101   |
| BIO-399   | BIO-101   |
| CS-190    | CS-101    |
| CS-315    | CS-101    |
| CS-319    | CS-101    |
| CS-347    | CS-101    |
| EE-181    | PHY-101   |

*prereq*

*result*

| course                     | prerequisite               |
|----------------------------|----------------------------|
| Intro. to Biology          | NULL                       |
| Genetics                   | Intro. to Biology          |
| Computational Biology      | Intro. to Biology          |
| Intro. to Computer Science | NULL                       |
| Game Design                | Intro. to Computer Science |
| Robotics                   | Intro. to Computer Science |
| Image Processing           | Intro. to Computer Science |
| Database System Concepts   | Intro. to Computer Science |
| Intro. to Digital Systems  | Physical Principles        |
| Investment Banking         | NULL                       |
| World History              | NULL                       |
| Music Video Production     | NULL                       |
| Physical Principles        | NULL                       |



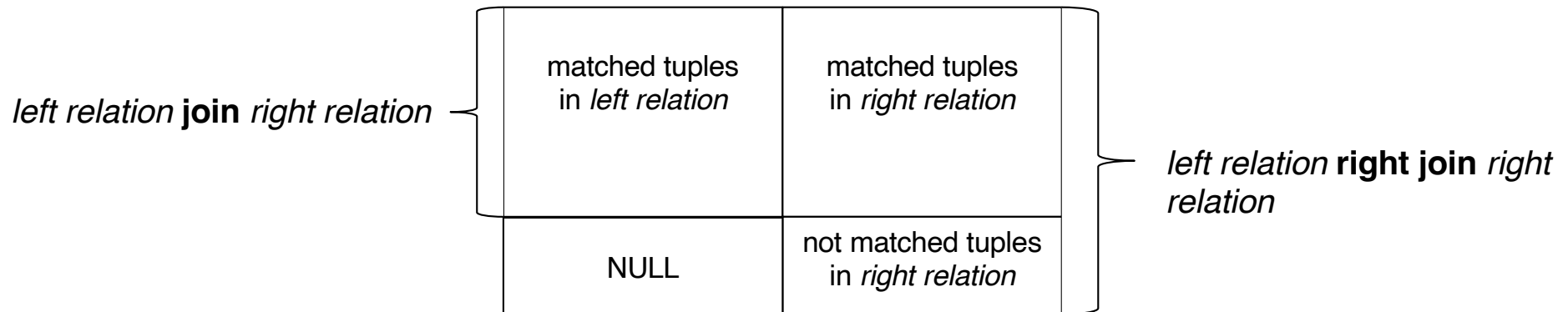
# Right Outer Join

- Computes the inner join and then adds tuples from the right relation that does not match tuples in the left relation to the result of the join.
- Find courses that are not prerequisite of any course

**select** *title*

**from** *prereq* **right join** *course* **as** *c* **on** *prereq\_id*= *c.course\_id*

**where** *prereq\_id* **is null**





## Right Outer Join (Cont.)

- Find all students who has no advisor.

**select** *name* **from** *advisor* **right join** *student* **on** *s\_ID* = *ID*  
**where** *i\_ID* **is null**

- Equivalent to:

**select** *name* **from** *student* **left join** *advisor* **on** *s\_ID* = *ID*  
**where** *i\_ID* **is null**

| <i>s_id</i> | <i>i_id</i> |
|-------------|-------------|
| 00128       | 45565       |
| 12345       | 10101       |
| 23121       | 76543       |
| 44553       | 22222       |
| 45678       | 22222       |
| 76543       | 45565       |
| 76653       | 98345       |
| 98765       | 98345       |
| 98988       | 76766       |

*advisor*

| <i>ID</i> | <i>name</i> | <i>dept_name</i> | <i>tot_cred</i> |
|-----------|-------------|------------------|-----------------|
| 00128     | Zhang       | Comp. Sci.       | 102             |
| 12345     | Shankar     | Comp. Sci.       | 32              |
| 19991     | Brandt      | History          | 80              |
| 23121     | Chavez      | Finance          | 110             |
| 44553     | Peltier     | Physics          | 56              |
| 45678     | Levy        | Physics          | 46              |
| 54321     | Williams    | Comp. Sci.       | 54              |
| 55739     | Sanchez     | Music            | 38              |
| 70557     | Snow        | Physics          | 0               |
| 76543     | Brown       | Comp. Sci.       | 58              |
| 76653     | Aoi         | Elec. Eng.       | 60              |
| 98765     | Bourikas    | Elec. Eng.       | 98              |
| 98988     | Tanaka      | Biology          | 120             |

*student*

| <i>name</i> |
|-------------|
| Brandt      |
| Williams    |
| Sanchez     |
| Snow        |

*result*



# Full Outer Join

- The **full outer join** is a combination of the left and right outer join
- Find courses that has no prerequisite and courses that are not prerequisite of any course

```
select c1.title as course, c2.title as prerequisite
from course as c1 full join prereq as p on c1.course_id=p.course_id
      full join course as c2 on c2.course_id=p.prereq_id
where c1.title is null or c2.title is null
```

